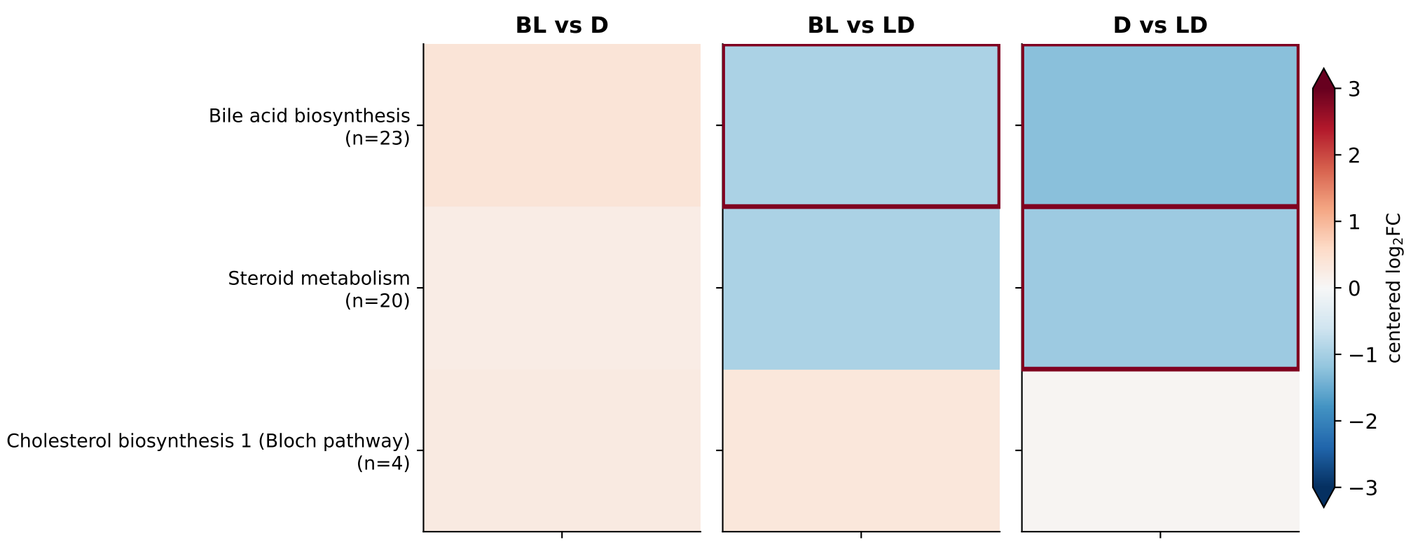
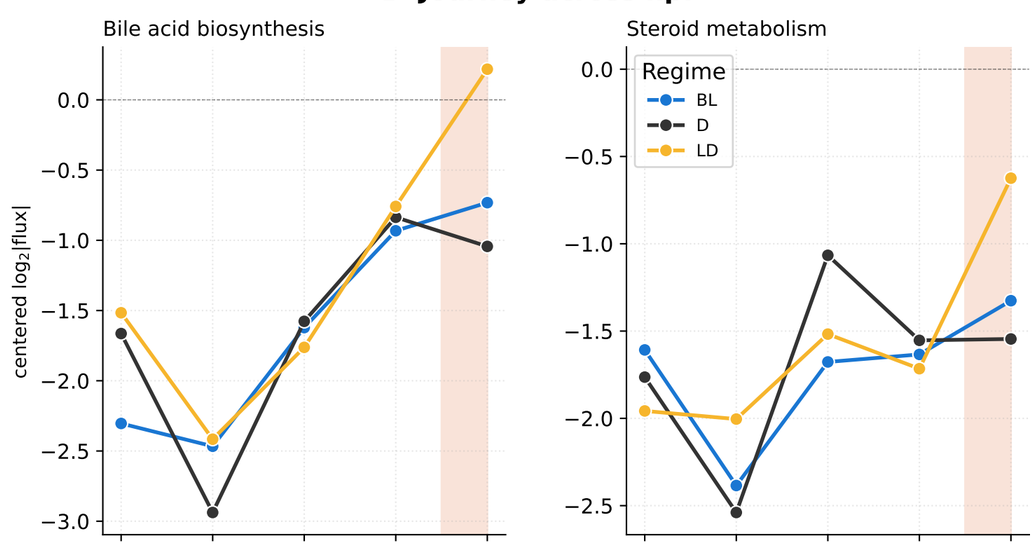
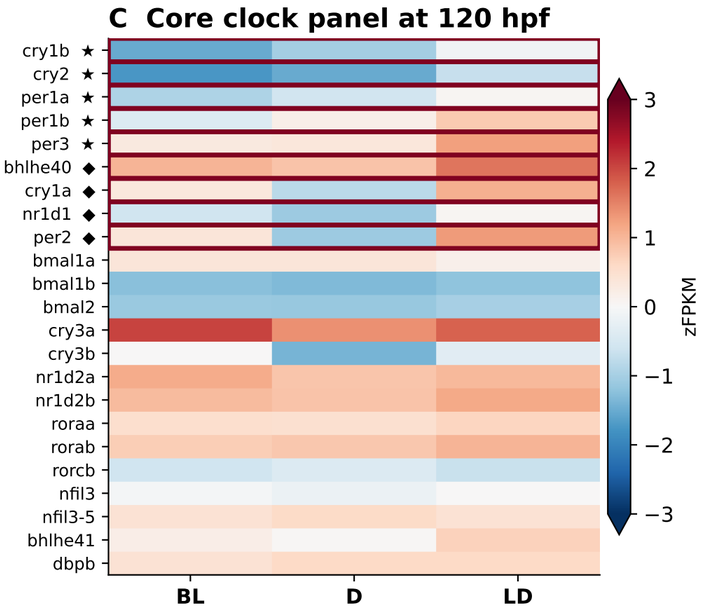
Figure 6 — LD alone steps up sterol / bile at 120 hpf, when the clock separates it from BL and D







D The two halves of Finding 6

Metabolism (panels A, B):

- At 120 hpf, LD alone steps up bile acid biosynthesis (23/23 reactions rise, $q = 1.2 \times 10^{-3}$) and steroid metabolism (18/18 reactions, $q = 2.2 \times 10^{-5}$)
- Flat in BL and D (96→120); cholesterol biosynthesis unchanged in all three
  → the step sits downstream of sterol synthesis, in sterol modification and bile acid formation
- At 120 hpf: LD > BL on bile acid (20/24 reactions, $q = 4.9 \times 10^{-11}$)
  LD > D on bile acid (19/21, $q = 1.7 \times 10^{-6}$) and on steroid (17/18, $q = 1.9 \times 10^{-3}$)
  BL and D do not differ on either subsystem

Clock transcripts (panel C):

- LD_120 highest on 17 of 23 genes with data
- On 9 of them LD exceeds both constants by ≥ 0.5 zFPKM (★ 5 BL ≤ D < LD; ◆ 4 D < BL < LD)
- No gene has D above BL by that margin

The metabolic step and the clock separation group the two constant regimes together against the cycling one — the sterol/bile step is the candidate metabolic consequence of entrained clock output.